

ORIGINAL ARTICLE

WILEY

Mechanisms regulating the airborne survival of *Klebsiella pneumoniae* under different relative humidity and temperature levels

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Funding information

None

Abstract

In this study, *Klebsiella pneumoniae* was suspended in synthetic saliva in a nebulizer (N_0) and nebulized for 5 min (N_5) into an aerosol chamber and further prolonged in the aerosolization phase for 15 min (A_{15}) under four different conditions: 20°C, 50% relative humidity (RH); 20°C, 80% RH; 30°C, 50% RH; and 30°C, 80% RH. Samples were collected at N_0 , N_5 , and A_{15} , then subjected to survival analysis and comparative transcriptomic analysis in order to help elucidate the underlying mechanisms of airborne survival. Survival analysis shows that a higher humidity and lower temperature were favorable for the airborne survival of *K. pneumoniae*, and the effect of RH was more remarkable at 20°C than that at 30°C. The RNA-seq results show that during the nebulization phase (N_0 vs. N_5), a total number of 201 differentially expressed genes (DEGs) were identified (103 downregulated and 98 upregulated). Comparison between nebulization and aerosolization phases (N_5 vs. A_{15}) indicates up to 132 DEGs, with 46 downregulated and 86 upregulated. The most notable groups of genes are those involved in cellular remodeling, metabolism and energy processes. Alarming, the *mbi* gene, which encodes antibiotic resistance in *K. pneumoniae*, was upregulated during the suspension phase under all the tested conditions. This study provides insights into the control of airborne transmitted diseases.

KEYWORDS

aeromicrobiology, airborne survival, antibiotic resistance, differential expression genes, *Klebsiella pneumoniae*, RNA-seq

1 | INTRODUCTION

Diverse natural and man-made processes could release bioaerosols. Research has been ongoing over the past 70 years, to investigate the survival of bioaerosols under various environmental conditions and the potential control measures to limit their dispersal.^{1–3} Bioaerosols play an important role in the transmission of airborne pathogens. Successful pathogen transmission is achieved when the infectious

agent from a source or index case arrives to a susceptible secondary host causing infection. Such infectious organisms (viruses, bacteria, or fungi) are exposed to a number of environmental hazards during this journey between hosts. Factors such as temperature, humidity (both relative and absolute), sunlight (ultraviolet light), and even atmospheric pollutants can all act to inactivate free-floating, airborne infectious organisms.⁴ These factors affect various infectious organisms in different ways and to different extents.

Air-acquired infections not only pose challenges to offices and commercial buildings, but also lead to a significant number of hospital infections.⁵ In particular, after the outbreak of COVID-19, the public has become more and more aware of the danger of airborne pathogens. Additionally, many of these bacterial and fungal pathogens have developed antibiotic resistance, thus putting further difficulties on treatment. The global public health leadership now believes that we are entering a “post-antibiotic era,” where once easily treated, infectious diseases will become increasingly difficult to treat.⁵ Although recommendations are in place to maintain hospital hygiene, they do not take the transmission of infectious airborne particles into consideration.⁶

Finding ways to control the transmission of bacterial aerosols can contribute to various applications such as infection control in hospitals and animal husbandry. Maintaining hospitals and other health-care premises at a particular RH and temperature may likely reduce the survival and therefore transmission of airborne bacteria.⁷ Such environmental recommendations are used in one particular hospital in Japan, wherein different locations are maintained at an average temperature of 24–27°C and relative humidity (RH) between 50–60% during the summer months and 20–24°C at 40–50% relative humidity during winter.⁶ However, these recommendations are mainly for human comfort rather than the control of infectious agents. If knowledge on the survival of pathogenic airborne bacteria under different environmental conditions (such as RH and temperature) can be obtained, building management teams may optimize the conditions maintained in the indoor environment to reduce airborne infection.

Although efforts are being made to contain the spread of infectious agents, the knowledge base is limited and translation to practice is in its infancy. Although attempts are made to investigate the effects of environmental parameters on the survival of microorganisms in air, limited data are available on the gene response of pathogens during airborne transit.⁷ Studying the gene expression profile may generate new knowledge about the genes important in bacterial airborne survival and thus help develop future engineering solutions to reduce airborne transmission of diseases. In this study, the survival and gene response of the Gram-negative bacterium *K. pneumoniae*, a common pathogen responsible for hospital acquired infections, under varying conditions of temperature and RH was investigated. The bacteria were nebulized into a chamber under varying conditions of temperature and humidity. Survival analysis and transcriptome sequencing were performed to determine the airborne viability and differentially expressed genes (DEGs) in the bacteria under the varying physical parameters.

2 | MATERIALS AND METHODS

2.1 | Bacterial strain

Fresh cultures of the bacteria *K. pneumoniae* were grown to an exponential phase in Luria Bertani medium (Affymetrix Inc., U.S.A.) in a shaker incubator at 37°C and 150 rpm. Bacteria were grown for

Practical Implications

- Understanding the underlying mechanisms that regulate bacterial survival in the air could help reduce disease transmission.
- A lower humidity and higher temperature may induce desiccation due to evaporation of the aerosol droplets, which could be beneficial in reducing the airborne transmission of *Klebsiella pneumoniae*.
- The upregulation of genes involved in antibiotic resistance to beta-lactam antibiotics during prolonged aerosolization under 20°C, 80% RH implies that airborne acquired infections may pose extra challenges for treatment in cool and moist climatic conditions.
- The present study highlights the importance of indoor conditions in disease prevention.

12 h to the late-log phase (10^9 CFU per ml, pre-determined by performing the growth curve) so as to mimic actively growing cells as observed during the course of an infection. The bacterial cells were harvested by centrifugation at 4,000 g for 6 min, then washed, and resuspended in artificial saliva (PBS+10% fetal calf albumin).⁸ The bacterial solution was then transferred to a six-jet Collison Nebulizer (BGI Inc., U.S.A.) for nebulization at 20 psi for 5 min.

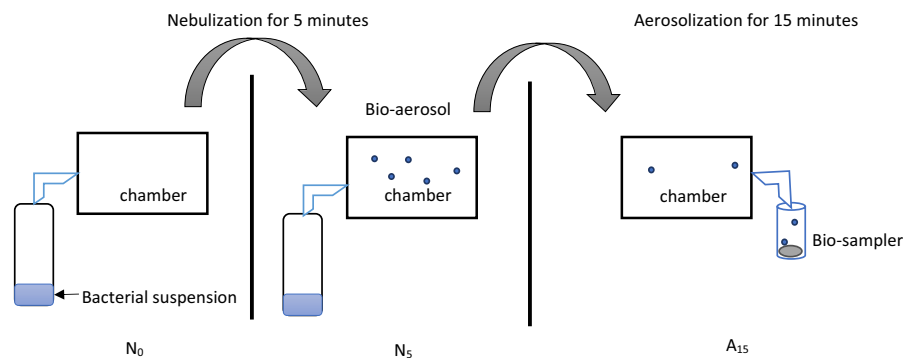
2.2 | Experimental setup

A standardized experimental setup that is commonly used in aerobiology was used in the present study (as shown in Figure 1).^{7,9–11} The Collison Nebulizer, commonly used for generating bioaerosol in animal infection and aerobiology studies, was used for the production of bioaerosols. During the nebulization process, compressed air was pumped into the nebulizer at a pressure level of 20 psi for 5 min to generate bioaerosols. The bioaerosols generated were ejected into a cylindrical bioaerosol chamber constructed by us (D: 50.8 cm × H: 58 cm) through a tunnel that connects the nebulizer and chamber. This bioaerosol chamber can be split in the middle to ensure that the inside of the chamber can be regularly cleaned to remove nucleic acid residues. The chamber and other aerosolization equipment and biosamplers were operated inside a Class II biosafety cabinet (Brand: Nuaire, Model: NU-543–600).

2.3 | Airborne survival under a combination of physical stresses

To test the survival of the bacteria, the bacterial suspension was nebulized for 5 min in a nebulizer, and the aerosols generated were ejected into the chamber for 15 min (Figure 1). The

FIGURE 1 A schematic diagram of the setup and different sampling points. Before nebulization, the bacteria suspended in the nebulizer were defined as N_0 . After 5 min of nebulization, the bacterial suspension was called N_5 . After completing the nebulization process, the aerosol was suspended in the air for 15 min before sampling as A_{15}



air temperature in the chamber was set at two varying temperatures, $22 \pm 2^\circ\text{C}$ and $30 \pm 2^\circ\text{C}$, and RH adjusted by either spraying sterile water or purging dehumidified air into the chamber to achieve RH conditions of 50% and 80%. The probe of a digital hygro-thermometer was inserted into the chamber to monitor the RH and temperature inside the chamber during the experiment. Both temperature and RH were kept constant during the course of an experiment. The bacterial culture was subjected to six different treatments: Treatment 1: before nebulization; Treatment 2: after nebulization; Treatment 3: suspension under condition 20°C , 50% RH; Treatment 4: suspension under condition 20°C , 80% RH; Treatment 5: suspension under condition 30°C , 50% RH; and Treatment 6: suspension under condition 30°C , 80% RH. Before nebulization, the bacteria suspended in the nebulizer were defined as N_0 . After 5 min of nebulization, the bacterial suspension was called N_5 . The bioaerosols further suspended in the air for 15 min were named A_{15} . Bacteria in the nebulizer (N_0 , N_5) and in the bacteria suspended in the air (A_{15}) were sampled at different sampling points as shown in Figure 1.

The culturable as well as the total bacterial counts were used to determine the log reduction in bacterial survival.⁷ The culturability of the bacterial cells was determined by the plate-count method with tryptone soy agar incubated at 37°C for 48 h and the DNA counts by quantitative polymerase chain reaction (qPCR). Bacterial DNA was extracted using a QIAamp DNA Mini Kit (Qiagen, Germany) following the manufacturer's protocol. The concentration of the extracted DNA samples was determined by qPCR using a QuantiNova™ SYBR® Green PCR Kit (Qiagen, Germany), with the forward primer *khe-F* (5'-GAT GAA ACG ACC TGA TTG CAT TC-3') and the reverse primer *khe-R* (5'-CCG GGC TGT CGG GAT AAG-3'). These primers bind to the hemolysin gene (*khe*), which is specific for *K. pneumoniae*.¹² The qPCR reaction was set up by first constituting a Master Mix solution according to the manufacturer's protocol, followed by aliquoting the Master Mix solution to each qPCR strip tube. Next, DNA templates were added into each strip tube containing aliquot of Master Mix solution. The thermocycling program of the AB StepOne RT-qPCR System (Applied Biosystems, Thermo Fisher, USA) consisted of an initial melt cycle at 95°C for 2 min, followed by 40 cycles of denaturation at 95°C for 5 s and annealing/extension at 60°C for 10 s. The *K. pneumoniae* culture was used to set a standard curve by performing a serial dilution of 10-folds.

Bacteria were sampled into a biosampler containing 5-ml PBS at a flow rate of 28 L/min for 3 min. The reduction in bacterial survival was calculated between different sampling points: nebulization (N_0 to N_5)—before and after nebulization; airborne suspension (N_5 to A_{15})—after 15 min of airborne suspension. Besides the bacterial survivability, transcriptome analysis for the different phases under the combination of varying physical parameters was also performed. All the experiments were performed in triplicates.

To account for the potential loss of bioaerosols during aerosolization and sampling, a normalized bacterial survival ratio was calculated as shown in Equation 1.

$$\text{Normalized bacterial survival ratio} = \frac{\text{number of culturable bacterial cells (plate - count data)}}{\text{total number of bacterial cells (qPCR data)}} \quad (1)$$

To assess the changes in the airborne bacterial survival, the log reduction between the normalized bacterial survival ratio during nebulization was calculated as: $\text{Log reduction} = \text{Log}_{10}N_5 - \text{Log}_{10}N_0$ (N_0 and N_5 are normalized bacterial survival ratios before nebulization and after nebulization for 5 min, respectively), and that after aerosolization was calculated as: $\text{Log reduction} = \text{Log}_{10}A_{15} - \text{Log}_{10}N_0$ (A_{15} is the normalized bacterial survival ratio after suspension in air for 15 min).

2.4 | RNA extraction and RNA sequencing

N_0 and N_5 samples were collected using RNa protect bacteria reagent at a reagent:sample ratio of 1:2 (v/v, total 0.9 ml). For the collection of the A_{15} airborne *K. pneumoniae* samples for RNA analysis, the bacterial cells were sampled into a biosampler containing 1:2 PBS: RNa protect bacteria reagent (v/v, total 20 ml) for 5 min. All the samples (N_0 , N_5 , A_{15}) in the RNa protect mixture were stored on ice immediately after collection, prior to RNA extraction.

To extract bacterial RNA, samples were centrifuged for 10 min at 5000 g. The supernatant was discarded, and the cell pellet was collected for RNA extraction using the RNeasy® Mini Kit (Qiagen, Germany) following the manufacturer's protocol. The concentration of the extracted RNA samples was determined using a Nanodrop system (NanoDrop, USA), and the integrity of the RNA was

examined by the RNA integrity number (RIN) using an Agilent 2100 bioanalyser (Agilent, USA). All samples with RIN values > 7.0 were used for the library construction. Sequencing was performed using the BGISEQ-500 sequencer platform (BGI Co. Ltd, China) by BGI Co. Ltd. The experiment was conducted in triplicates for comparative transcriptome analysis.

2.5 | Bioinformatics analysis

Followed by RNA-seq, bioinformatic analysis was also conducted by BGI Co. Ltd. The scope of the study was to determine the change in gene expression patterns during nebulization and suspension under the four combinations of temperature and humidity (22°C, 50% RH; 22°C, 80% RH; 30°C, 50% RH; and 30°C, 80% RH). At least 19 Mb clean reads were obtained for each sample (Table S1) and then mapped to *K. pneumoniae subs. Pneumoniae WGLW2* genome (NCBI accession: NZ_AMLM000000000) using HISAT/Bowtie2 tool.^{13,14} Biological replicates within each treatment groups demonstrate >0.9 correlation (Pearson value), indicating high reproducibility of replicates (Figure S1).

Next, the gene expression level was quantified by RSEM software package.¹⁵ RNA expression of *K. pneumoniae* in the different phases and under the different conditions was compared pairwise using NOISeq method to determine the DEGs.¹⁶ DEGs were defined as |fold change| >2 and probability >0.8. The DEGs were subjected to annotation analysis of gene ontology (GO) and the GO functional classification using WEGO software.¹⁷ Principal component analysis was performed for all the samples (Figures S2 and S3).¹⁸

3 | RESULTS

3.1 | Survival of *K. pneumoniae* under varying physical parameters

Figure 2 depicts a low log reduction in survival following nebulization, showing that the process of nebulization is not detrimental to the bacterial cells. Among the four conditions tested (20°C 50%, 20°C 80%, 30°C 50%, and 30°C 80%), the maximum death (nearly 4-log reduction in survival) of the organism was observed under 50% RH at 30°C, a dehydrating condition. Under the conditions of 80% RH and 20°C, the airborne survival was maximum with a log reduction of 0.6.

The results obtained from the one-way ANOVA statistical analysis show that there is a significant difference in survival of the organism between the different conditions tested (Table S2). The results also point out that at an increased temperature of 30°C ($p = 0.788$), the effect of RH on death of the organism is not as profound as at 20°C ($p = 0.027$). In addition, the death observed at 20°C, 50%RH is not significantly different from that observed at 30°C, 80% RH ($p = 0.061$). It can therefore be stated that at lower temperatures, RH will significantly impact organism survival.

However, the effect of RH is reduced at higher temperature and temperature appeared to impact bacterial survival more significantly compared with RH.

3.2 | Differentially expressed genes in the nebulization phase

The DEGs identified in the nebulization phase cover three GO terms (cellular process, biological process, and molecular function) in the KEGG pathway analysis (Figure 3). Among the genes in these metabolic pathways, the operon for the degradation of phenyl acetic acid was highly upregulated (Table S3). In addition, the genes *nuoL* and *nuoN* were also considerably upregulated. These genes encode the protein NADH-quinone oxidoreductase subunit L and NADH-quinone oxidoreductase subunit N, respectively, which are involved in ATP synthesis-coupled electron transport. This could be indicative of the organism preparing for the airborne suspension phase by the increased production of energy required for further metabolic process. Another interesting finding is the upregulation of the fatty acid oxidation genes *FadB* and *FadA*. The gene *entF* encoding enterobactin synthase subunit F was also found to be upregulated. The possible role of this gene could be that of iron sequestration as iron is essential for the survival of organisms, also involved in the electron transport chain in the Fe-S centers. Siderophore-mediated iron sequestration is an additional virulence determinant in *Klebsiella* species.

Among the downregulated genes (Table S4), a large number of genes comprised those involved in cellular processes. Genes encoding for proteins like peptidoglycan-binding protein LysM, lipid A biosynthesis, palmitoleoyl acyl transferase, and fimbrial proteins were downregulated. Among the metabolic processes, genes encoding proteins for ethanolamine utilization were downregulated. Molecular chaperone protein encoding genes were also found to be downregulated during the nebulization phase.

3.3 | Differentially expressed genes in the airborne suspension phase under the varying conditions of temperature and humidity

The gene expression under the varying conditions of temperature and humidity shows a close correlation (Figure S1). The cluster hierarchy and PCA plots reveal that the four treatments cluster together while the treatments before nebulization and after nebulization are clustered into separate groups (Figures S2 and S3). The DEGs identified in the aerosolization phase cover three GO terms (cellular process, biological process, and molecular function) (Figures 4-7). The commonly upregulated genes across all the four treatments (20°C 50%, 20°C 80%, 30°C 50%, and 30°C 80%) in biological process comprised genes encoding proteins involved in ethanolamine utilization (*eut*), microcompartment formation, and maltose/maltodextrin uptake and utilization. Ethanolamine catabolism requires the

FIGURE 2 The log reduction in survival of *K. pneumoniae* under four different conditions of temperature and RH. Error bars represent the standard deviation of replicates ($n \geq 3$)

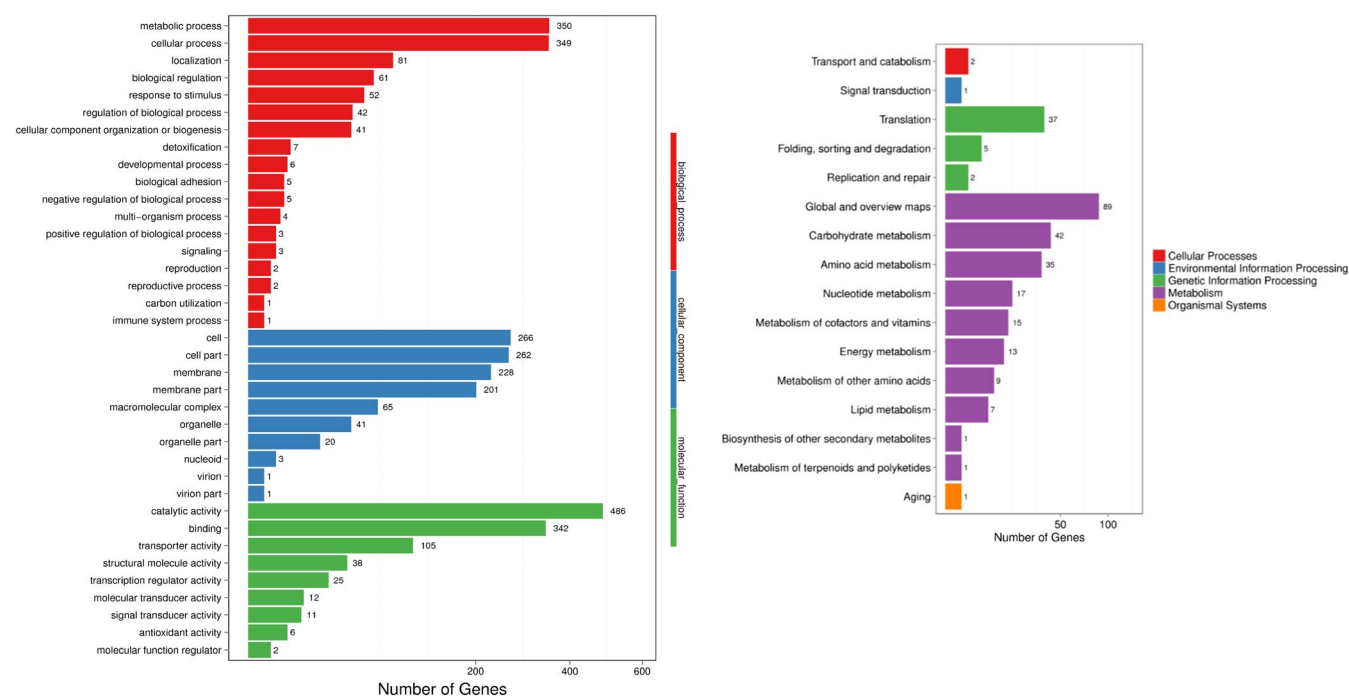
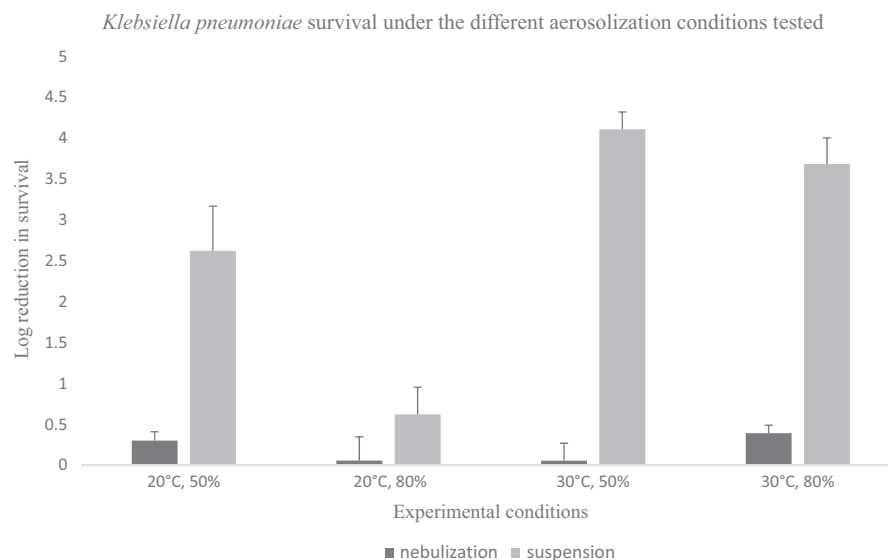


FIGURE 3 Bar Plots summarizing the gene ontology terms over-expressed in differentially expressed genes as well as the KEGG pathway functional enrichment analysis of significant DEGs before and after nebulization (N_0 vs. N_5 ; treatment 1 vs. treatment 2). The DEGs identified cover three GO terms: cellular component, biological process and molecular function. The vertical axis represents the GO term or KEGG pathway terms significantly increased by the DEGs; the horizontal axis indicates the number of genes. GO gene ontology, DEG differentially expressed gene

construction of bacterial microcompartments that serve to concentrate reaction intermediates and potentially limit toxic effects of acetaldehyde on the cell.

The cold shock protein CspC was upregulated across all the treatments (Table S5); this could be attributed to the fall in temperature due to droplet evaporation in addition to the lower temperatures maintained within the chamber as compared to that during nebulization (37°C). During nebulization and suspension under 20°C, 80% RH, DEGs clustered into pathways conferring antibiotic resistance

to the pathogen (Figure S6). Among the 20 pathways enriched in KEGG analysis were the pathways for biosynthesis of antibiotics, biosynthesis of secondary metabolites, and carbapenem biosynthesis. Carbapenemase-producing *K. pneumoniae* could pose a serious threat in nosocomial infections as carbapenems are used as the last line of drug treatment against clinical isolates. The gene *mbi* encoding metallo-beta-lactamase protein was found to be upregulated during aerosolization under all the conditions tested in this study. Activities carried out by members of this superfamily include class B

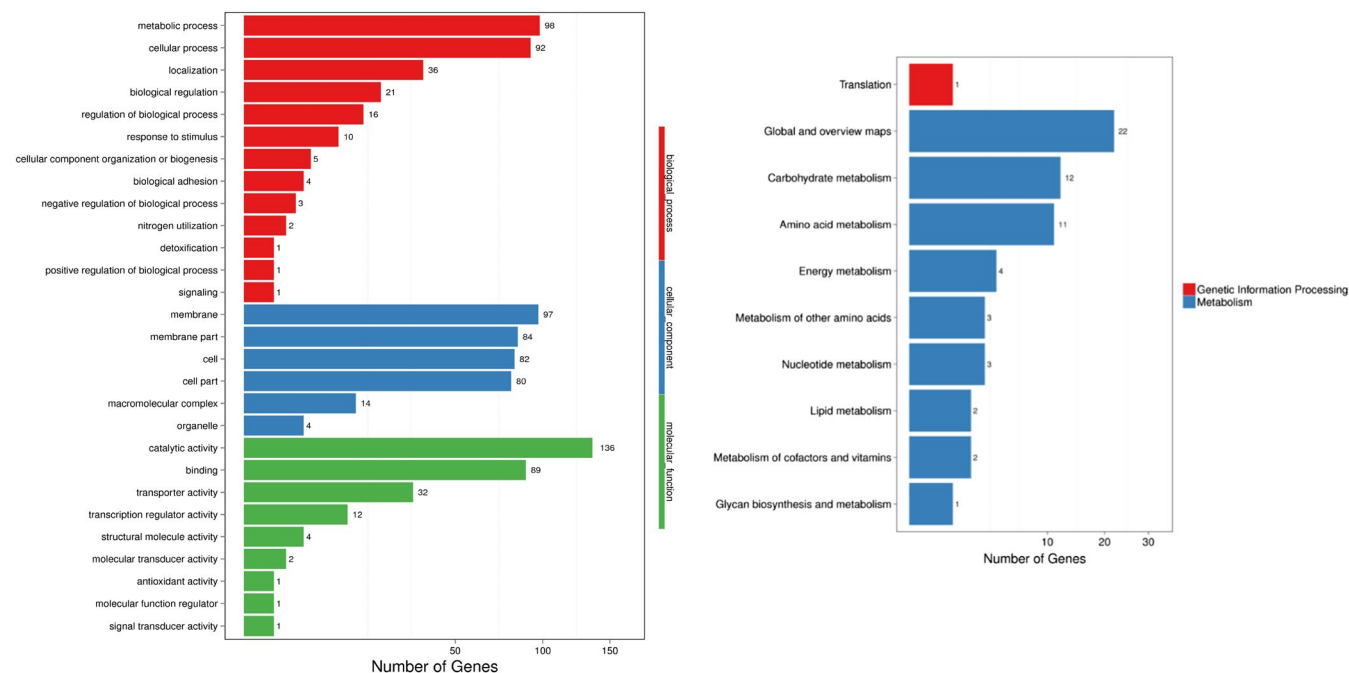


FIGURE 4 Bar Plots summarizing the gene ontology terms over-expressed in differentially expressed genes as well as the KEGG pathway functional enrichment analysis of significant DEGs during aerosolization under condition 20°C, 50% RH (N_5 vs. A_{15} ; treatment 2 vs. treatment 3). The DEGs identified cover three GO terms: cellular component, biological process and molecular function. The vertical axis represents the GO term or KEGG pathway terms significantly increased by the DEGs; the horizontal axis indicates the number of genes. GO gene ontology, DEG differentially expressed gene

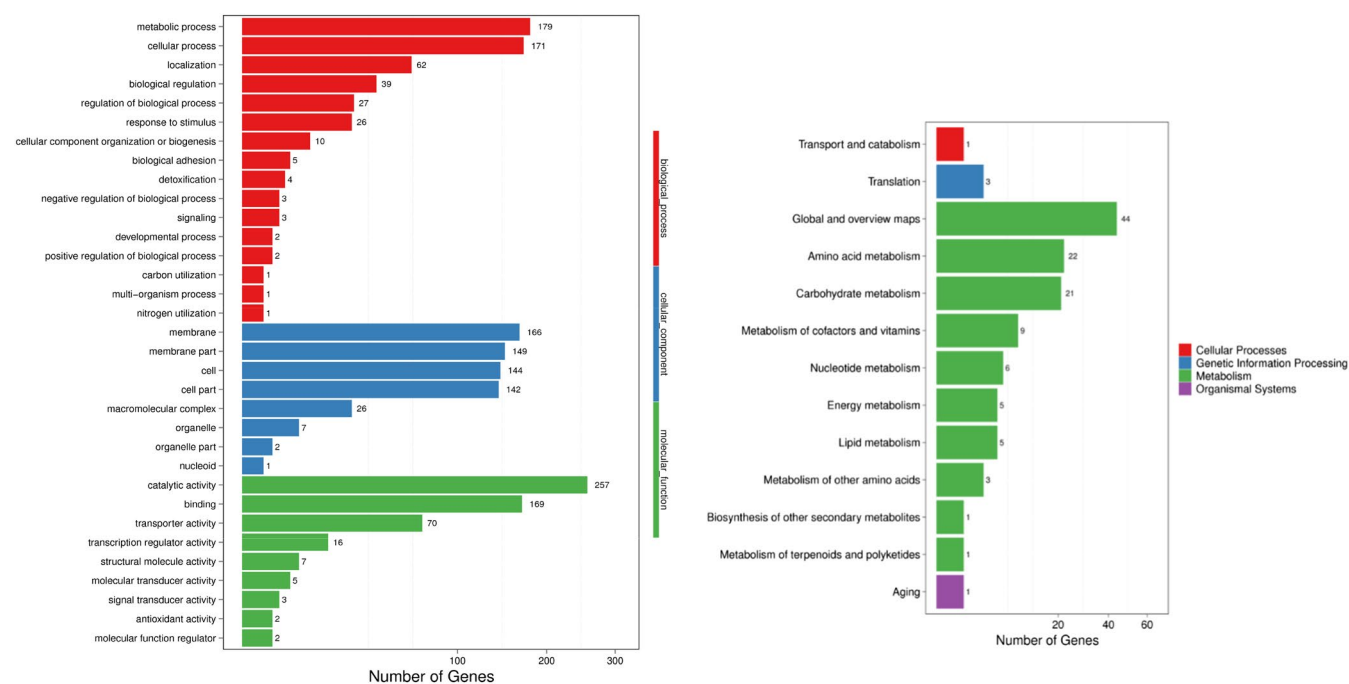


FIGURE 5 Bar Plots summarizing the gene ontology terms over-expressed in differentially expressed genes as well as the KEGG pathway functional enrichment analysis of significant DEGs during aerosolization under condition 20°C, 80% RH (N_5 vs. A_{15} ; treatment 2 vs. treatment 4). The DEGs identified cover three GO terms: cellular process, biological process and molecular function. The vertical axis represents the GO term or KEGG pathway terms significantly increased by the DEGs; the horizontal axis indicates the number of genes. GO gene ontology, DEG differentially expressed gene

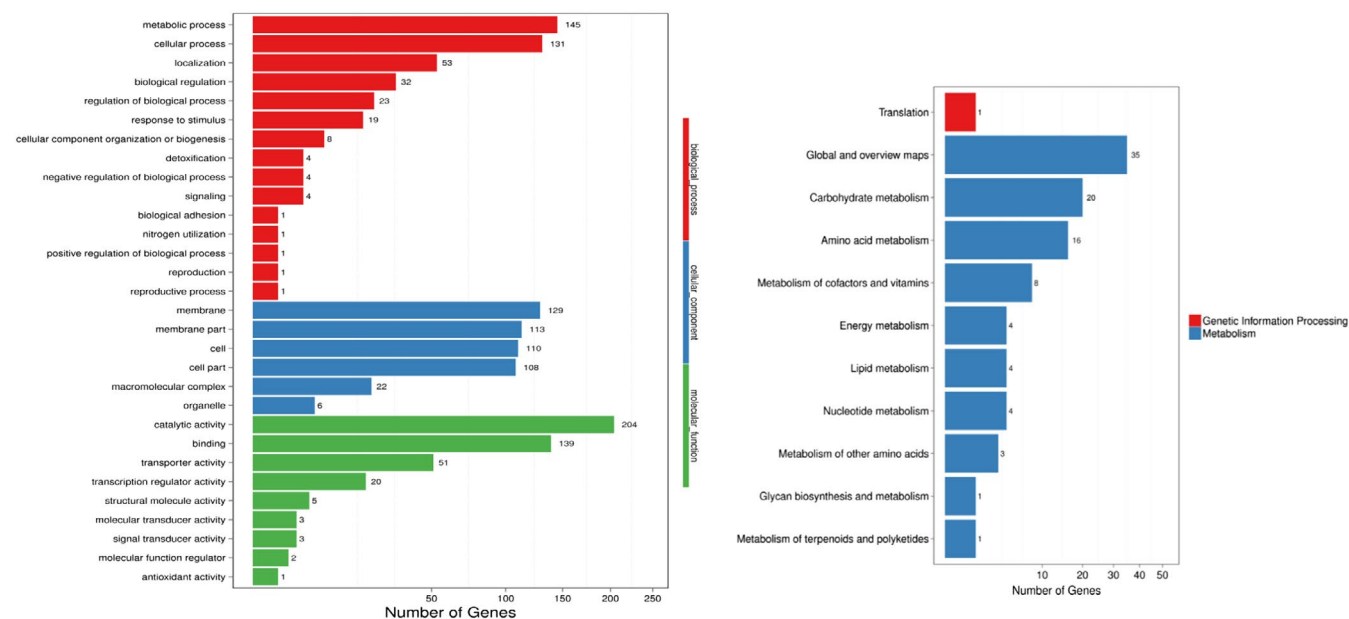


FIGURE 6 Bar Plots summarizing the gene ontology terms over-expressed in differentially expressed genes as well as the KEGG pathway functional enrichment analysis of significant DEGs during aerosolization under condition 30°C, 50% RH (N_5 vs. A_{15} ; treatment 2 vs. treatment 5). The DEGs identified cover three GO terms: cellular process, biological process and molecular function. The vertical axis represents the GO term or KEGG pathway terms significantly increased by the DEGs; the horizontal axis indicates the number of genes. GO gene ontology, DEG differentially expressed gene

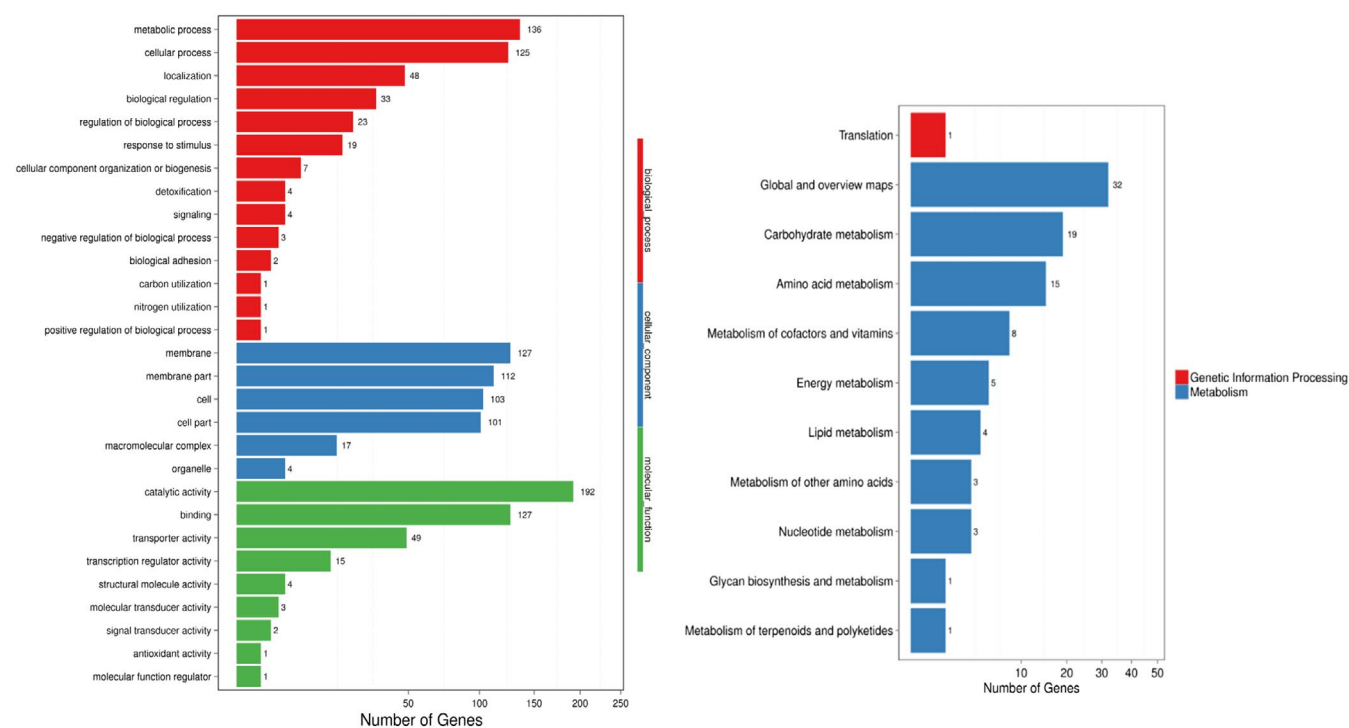


FIGURE 7 Bar Plots summarizing the gene ontology terms over-expressed in differentially expressed genes as well as the KEGG pathway functional enrichment analysis of significant DEGs during aerosolization under condition 30°C, 80% RH (N_5 vs. A_{15} ; treatment 2 vs. treatment 6). The DEGs identified cover three GO terms: cellular process, biological process and molecular function. The vertical axis represents the GO term or KEGG pathway terms significantly increased by the DEGs; the horizontal axis indicates the number of genes. GO gene ontology, DEG differentially expressed gene

beta-lactamases, which can catalyze the hydrolysis of a wide range of beta-lactam antibiotics. This is an attractive finding, since the expression of the *mbi* gene in the aerosolization phase could point towards the preparation of airborne bacteria to successfully evade antibiotic treatment and cause an infection in the host.

Among the cellular process, the genes encoding peptidoglycan-binding protein LysM, lipid A biosynthesis palmitoleoyl acyl transferase and the periplasmic protein CpxP were upregulated. A unique feature to the treatment 4 (20°C, 80% RH) is the expression of the gene encoding the multidrug transporter subunit MdtJ.

Furthermore, a large number of genes encoding for proteins belonging to the ABC transporter superfamily and MFS transporter superfamily, which were expressed during the nebulization phase, are downregulated during the suspension phase (Table S6). This downregulation of transmembrane transport porins results in a decrease in outer membrane permeability, restricting the entry of metabolites or even drugs into the cell, which may confer antibiotic resistance to the bacterial cell. A noteworthy mention is the downregulation of the genes *marR*, *marA*, and *marB*. The *marRAB* operon is a dedicated operon that confers multiple antibiotic resistance to Gram-negative bacteria by increasing drug efflux and decreasing influx of the antibiotic through the Outer Membrane Protein F (*ompF*).¹⁹ Finally, the genes involved in phenyl acetic acid metabolism are downregulated in the airborne phase as opposed to the nebulization phase.

4 | DISCUSSION

The complete set of RNA transcripts produced by the genome (the transcriptome) was studied using high-throughput methods to obtain information about the genes that are differentially expressed in distinct cell populations, or in response to different treatments. In this study, comparative transcriptome analysis was used to determine the differentially expressed genes in *K. pneumoniae* during aerosolization from liquid suspension under varying environmental conditions. This method could help screen and deduce the potential stimuli and stressors that the bacteria encounter, as well as the potential survival adaptations, in response to the changes in the external environment by tracing specific stress-inducible and stress-associated genes.^{20,21} In the study of the airborne transmission and control of bacterial aerosols, this information is valuable because identifying the environmental stressors and the bacterial adaptations in response to the new environment could help to predict the transmission risk of different bacteria under various environmental conditions and inspire researchers to develop new infection control technology.

In this work, with obtaining at least 19 Mb of clean reads from each sample, we analyzed the DEGs in *K. pneumoniae* during the nebulization phase (N_0 vs. N_5) and transition from the liquid to airborne phase under various temperature and RH conditions (N_5 vs. A_{15}). A total number of 201 (103 downregulated and 98 upregulated) and 132 DEGs (46 downregulated and 86 upregulated) were found for these two groups of comparison, respectively (Tables S3–S6). The

most notable group of genes are those involved in cellular remodeling, metabolism, and energy processes. To the best of our knowledge, this is the first study to employ high-throughput sequencing technique to analyze the transcriptomic changes in airborne bacteria in response to varying environmental conditions.

4.1 | Survival of *K. pneumoniae* under varying physical parameters

The results depict that when the temperature increased from 20°C ± 2 to 30°C ± 2, an overall decrease in the survival of *K. pneumoniae* could be observed. This observation is in agreement with previous studies, which showed an increase in death of airborne bacteria on increasing the temperature above 24°C.⁴ This trend has been found with members of Gram-negative, Gram-positive, and intracellular bacteria: *Pseudomonas* sp.,²² *Pasteurella* sp.,²³ *Salmonella* sp.,²⁴ *Serratia* sp.,²³ *Escherichia* sp.,^{23,24} *Bacillus* sp.,²³ *Bordetella* sp.,⁴ *Chlamydia* sp., and *Mycoplasma*.^{25,26}

The results obtained in this study depict a significant effect of RH on the survivability of *K. pneumoniae*. Previous studies on airborne Gram-negative bacteria such as *Serratia marcescens*, *Escherichia coli*, *Salmonella pullorum*, *Salmonella derby*, *Pseudomonas aeruginosa*, and *Proteus vulgaris* have reported an increase in death rates at intermediate (approx. 50–70%) to high (approx. 70–90%) RH environments.^{27,28} In contrast, another study on the Gram-negative bacillus, *K. pneumoniae*, demonstrated the relative stability at an intermediate RH of 60%.²⁹ Furthermore, another experiment with the Gram-negative rod *Pasteurella* species showed a greater survival in aerosols at higher RH levels.^{24,30} Survival of *Enterobacter cloacae*, *Erwinia herbicola*, *Klebsiella planticola*, and *Pseudomonas syringae* was highest at high relative humidity (70–80%) and low temperatures when sprayed in a greenhouse.³¹

Additionally, the results depict an increase in death under a shift towards a more dehydrating environment. This can be explained by the physical models described by Parienta et al. (2011), wherein³² water contained in droplets evaporates quickly within milliseconds thus reducing the temperature and increasing the solute content around the bacteria imposing additional osmotic stress to the bacteria. Moreover, Beney et al. (2004) showed that osmotic dehydration prompted a transition of membrane phospholipids from lamellar liquid crystal to lamellar gel in *E. coli*.³³ Therefore, the bacterial death due to dehydration can be attributed to a combination effect of membrane deformation and structural changes of the membrane lipids.

4.2 | Genes involved in metabolic processes

The observed upregulation of *eut* genes, genes encoding ethanolamine (EA) utilization, during the suspension phase may be advantageous to the bacterium upon host entry and colonization, which implies that airborne pathogens may pose an increased threat to

humans. Phosphatidylethanolamine is an abundant phospholipid in both mammalian and bacterial cell membranes, which can be readily broken down into glycerol and ethanolamine by phosphodiesterases. Ethanolamine is further split into acetaldehyde and ammonia. The ammonia can serve as a cellular supply of reduced nitrogen, and the acetaldehyde can be converted into the metabolically useful compound acetyl CoA, which then enters the TCA pathway, to produce the energy-yielding molecule ATP,³⁴ which is upregulated under all the environmental conditions tested (Figures S5–S8). Many gut-associated bacteria such as *Escherichia*, *Listeria*, *Clostridium*, *Salmonella*, *Enterococcus*, and *Klebsiella* contain genes required for the catabolism of EA.³⁴ There has been growing evidence of the association of *eut* genes with organism pathogenesis. *Eut* gene expression was significantly upregulated in *Listeria monocytogenes* in the intestines of mice as well as human epithelial colon cells.³⁵ Additionally, *eutB* deletion mutant was defective in intracellular replication and attained a log lower CFU when compared to the wild-type strain.³⁶ An intravenous mouse model infected with the mutant strain showed a decrease in CFU in the spleen and liver of the mouse.³⁷ EA is contained in the membrane lipid phosphatidyl ethanolamine, which is the second most abundant phospholipid in mammalian cells.³⁸ Studies revealed the preferential attachment of enterohemorrhagic *E. coli* and enteropathogenic *E. coli* to phosphatidylethanolamine compared with phosphatidylcholine and phosphatidylserine demonstrating a possible role of EA in the adherence of pathogens to the human epithelium cells.³⁹ Recently, the evidence of the role of ethanolamine during urinary tract infections has also been demonstrated.⁴⁰ There is also evidence for a correlation between disease progression and EA catabolism related to periodontal disease.⁴¹

Strikingly, the genes involved in fatty acid metabolism and biosynthesis are differentially regulated during nebulization as well as under the varying conditions of airborne suspension. These genes (*fadA* and *fadB*) are involved in the beta-oxidation pathway for the degradation of saturated fatty acids. For the degradation of unsaturated fatty acids, *fadH* gene products are involved.⁴² As such, it can be envisaged that the saturated fatty acids may be degraded as a source of energy in addition to maintaining membrane homeostasis under aerosolization stress. The fluidity of bacterial membranes is modified in response to several environmental cues, enabling survival in otherwise unfavorable conditions. One of the most important parameters affecting fluidity and rigidity of membranes in most bacteria is the ratio of unsaturated fatty acids to saturated fatty acids.⁴³ A study on the cold adaptation (20°C) in *Bacillus subtilis* depicted a change in the fatty acid composition.⁴⁴ Another study on *Cryptococcus flavescens* showed alterations in membrane as a response to cold adaptation.⁴⁵ Furthermore, another response to cold and presence of solvents is a change in the head groups of phospholipids. Cold adaptation in an *Arthrobacter* species was shown to suppress the synthesis of charged lipids and reduce the membrane lipid composition to only lipids with glycerol head groups.⁴⁶ Studies on effects of organic solvents revealed that in an adaptive mechanism of solvent-tolerant bacteria, the amount of phosphatidyl glycerol

relative to phosphatidyl ethanolamine in the membrane changes.⁴⁷ As such, the increased expression of the *eut* genes during aerosolization may also be involved in the remodeling of lipid head groups in order to maintain the stability and order of the membrane while lowering the permeability.⁴⁸

4.3 | Genes associated with nutrient starvation

The expression of the maltose system in the suspension phase is a sign of nutrient starvation as this system is expressed under conditions of glucose limitation via the increased synthesis of the inducer maltotriose.^{49,50} In addition, the genes associated with the β oxidation pathway are upregulated under all the conditions tested during airborne suspension (Figures S5–S8) thus indicative of a nutrient-starved environment. This meets the expectation that the suspending medium as well as the environment does not provide much nutrient to the bacteria. Furthermore, a drier environment (50% RH) induces a severe condition of starvation as the pathway for gluconeogenesis is upregulated under these environments as seen in treatments 3 (20°C, 50% RH) and 5 (30°C, 50% RH; Figures S5 and S7).

Genes involved in iron sequestration (*ent*) are upregulated during the nebulization phase. This could be explained by previous findings, wherein enhanced siderophore production reduced the level of oxidative stress experienced due to dehydration in *Klebsiella*.⁵¹ Additionally, genes (*dps*) encoding an Fe-binding and storage protein were significantly upregulated during aerosolization and *dps* mutants showed considerably lower survival upon aerosolization as compared to the parent strain.¹⁰ Thus, inferring siderophore production could play an important role during bacterial aerosolization and prolonged survival by counteracting the oxidative stress experienced due to dehydration (drying of the aerosol particles) during aerosolization.

In addition, siderophores contribute to the virulence of a wide variety of bacterial pathogens,⁵² including *K. pneumoniae*.^{53,54} High-affinity iron chelators or siderophores, such as aerobactin, enterobactin, salmochelin, and yersiniabactin, enable *K. pneumoniae* to cope with scarcity of iron in the mammalian hosts.⁵⁵ Among the different molecules, enterobactin has the highest known affinity for iron.⁵⁶ Enterobactin appears to play a dominant role in iron acquisition under iron-limiting conditions in vitro when compared with *in vivo* using intranasal infection model.⁵⁴ Additionally, enterobactin has also been associated with the ability to reduce oxidative stress in *E. coli* and inhibits myeloperoxide-enhancing *E. coli* survival in the inflamed gut.^{57,58}

4.4 | Genes comprising cellular processes

The genes involved in the pathways for the metabolism of sphingolipids and biosynthesis of glycosphingolipids were upregulated under treatments 3, 5, and 6 (Figures S5, S7 and S8). The properties of the lipid bilayer are largely influenced by the presence of

sphingolipids and glycosphingolipids, which have a higher melting temperature and thus reduce membrane fluidity. This can be attributed to the hyperosmotic stress experienced by the bacteria under the dehydrating conditions due to the rapid drying of the aerosol droplet. The pathways for the metabolism of sphingolipids and biosynthesis of glycosphingolipids were not upregulated under the moist conditions tested in treatment 4 (20°C, 80% RH), which seemed to be a more favorable environment for the airborne survival of the bacterium. The fluidity of membranes is modified in response to several environmental cues, enabling bacterial survival in otherwise unfavorable conditions.⁴³ For example, bacteria increase their membrane fluidity to adapt to low temperatures and hydrostatic pressures, while they decrease membrane fluidity, by densely packed saturated fatty acid containing phospholipids, in response to osmotic stress. The maintenance of bacterial membrane fluidity plays an important role in a variety of cell physiological functions such as nutrient transport, protection from external adverse environments, and cell morphology. Thus inferring, alterations in cellular membrane composition and structure are proven to be a crucial survival mechanism in bacterial pathogens and could also play a significant role in the survival of airborne bacterial pathogens under varying environmental conditions.

Interestingly, genes associated with modification to lipid A were upregulated under all conditions in the suspension phase. These genes catalyze the transfer of palmitoleate from palmitoleoyl acyl carrier protein to Kdo₂-lipid IV(A) to form Kdo₂-(palmitoleoyl)-lipid IV(A). Such modifications to lipid A have shown to provide resistance to stresses, such as harsh pH, low temperature, and antimicrobial peptides (APs).⁵⁹ APs are directed to LPS on the bacterial surface by electrostatic interactions. Colistin (polymyxin E) and polymyxin B are two antimicrobials, which share the interaction with the anionic LPS with mammalian APs.⁶⁰ Colistin is one of the last-line antimicrobials used to treat infections caused by multidrug-resistant strains of Gram-negative bacteria.⁶¹ *Klebsiella pneumoniae* may exhibit such a connection between Lipid A modifications and antibiotic resistance.⁶²

Additionally, the lipid A moiety of LPS triggers the mammalian innate immune system via the TLR4/MD2 receptors and induces macrophages,^{63,64} at picomolar levels, to synthesize potent mediators of inflammation like TNF- α and IL-1 β .⁶⁵ Modifications to lipid A may aid pathogens to modulate the activation of host defense mechanisms by altering the activation of the Toll-like receptor 4 (TLR4)/MD2 receptor complex.⁶⁶ Furthermore, the addition of palmitoleoyl to lipid A increases the unsaturation of lipids, which is a response associated with cold stress, and mutants defective in this process showed an increased susceptibility to antibiotics at low temperatures due to the impaired fluidity. This adaptive feature has thus far only been seen in bacteria that are capable of surviving outside the host and not the bacteria that depend on direct host-host transfer.

Another membrane component upregulated under all the conditions during suspension phase is the periplasmic protein cpxP encoded by the *cpxP* gene. This gene is upregulated during the cpx envelop stress response. The elevated expression of Cpx-regulated

factors is necessary to maintain proper protein folding in the bacterial envelope (inner membrane, periplasm, and outer membrane) and thus maintain the integrity of the cell.⁶⁷ In addition, Cpx envelope stress response has also been suggested to play an important role in the virulence of pathogenic organisms.^{68,69}

The component of the cell membrane, type 1 fimbrial protein, is downregulated during nebulization. Fimbriae (also known as Pili) mediate the process of adhesion of the bacterial pathogen to the host mucosal surfaces, which is essential in the establishment of infection and biofilm formation.^{70,71} Hence, the downregulation of this protein prompts to the idea that the bacteria sense the expulsion into the air are preparing for extended airborne suspension and further transmission. As expression of *pili* during airborne transmission could also result in attachment to inanimate surfaces and thus limit their dispersal. Fimbriae do not affect the ability of *K. pneumoniae* to colonize the intestine or to infect the lung,⁷² and thus, this may be a structural modification to ensure transmission and dispersal of the pathogen for a prolonged period and over greater distances from the source. All these instances confirm the hypothesis that these pathogens "sense" their ejection into the environment and prepare for the prolonged airborne survival.

4.5 | Environmental implications

Klebsiella pneumoniae is a pathogenic bacterium causing various infections in human. It is associated with hospital infections and contagious particularly to immune-compromised patients. The RH and temperature levels examined in the present study, 20 and 30°C, 50 and 80% RH, are typical indoor conditions when air-conditioning systems are in operation (20°C, 50% RH) and off (atmospheric level, 30°C, 80% RH) in tropical and subtropical areas like Hong Kong. Our results imply that higher temperature and lower RH would likely better reduce the viability of airborne pathogens. 50% RH has also been demonstrated to be ideal for inhibiting viral viability, including SARS-CoV-II.^{73,74} Therefore, it may be feasible to reduce airborne transmission of pathogens by maintaining a lower RH and relatively higher temperature within the range of comfortable hygrothermal conditions. Notably, although generation of aerosols using nebulizers is a widely adopted experimental setup in aerobiology field, this process can be quite different from human coughing and sneezing. Therefore, further research is needed to evaluate whether the data produced by nebulizers are extrapolatable to real situations.

Thus far, indoor management in most places in the world, such as the Indoor Air Quality (IAQ) Scheme in Hong Kong, only considers human comfort. Our current study stresses the importance of indoor hygrothermal conditions in infection control and simultaneously sheds light on the development of novel engineering solutions for controlling air-transmitted diseases.

Alarming, genes like *mbi* and carbapenem resistance genes, involved in antibiotic resistance in *K. pneumoniae*, were upregulated during airborne suspension. The exposure to or the infections caused by bacteria conferring antibiotic resistance could pose extra

challenges on treatment. *Klebsiella pneumoniae* is considered one of the major components of a group of multidrug-resistant ESKAPE (*Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter* spp) pathogens responsible for serious healthcare-associated infections. Therefore, the upregulation of antibiotic resistance genes during airborne transmission may alert a severe situation as compared to other exposure pathways.

5 | CONCLUSIONS

Understanding the underlying mechanisms that govern the survival of pathogenic bacteria in the aerosol could help reduce airborne transmission of diseases. Here, we present the DEGs associated with the survival of *K. pneumoniae* during airborne suspension, suggesting the important role of cellular remodeling and metabolic pathways in regulating bacterial survival in the air. A down-regulation of the structural attachment proteins like the fimbrial protein could be suggestive of the pathogen's predisposition to ensure its dispersal and "free movement" during the airborne phase. Furthermore, genes involved in energy processes were upregulated during nebulization (N_5). Also, there was an upregulation in the anti-apoptotic BAX protein during aerosolization (A_{15}). Both of these findings together are indicative of the pathogen's preparedness to endure and survive in the airborne phase. Strikingly, an upregulation in the *mbi* gene could pose serious challenges during prophylactic antibiotic measures in the spread of nosocomial infections. Furthermore, in addition to the upregulation of *mbi* gene, carbapenem resistance genes are upregulated under 20°C and 80% RH conditions, which points to an added advantage of the bacterium against the commonly used antibiotic regimes. This could result in a delayed treatment of the infection under cool and moist environmental conditions (such conditions are commonly experienced during spring or monsoon). Our findings may provide scientific basis for developing engineering solutions to prevent air-transmitted diseases and antibiotic resistance.

ACKNOWLEDGMENT

The authors would like to thank Dr Ka Man Lai for her expert advice on this study.

CONFLICT OF INTEREST

No conflict of interest declared.

AUTHOR CONTRIBUTIONS

Natasha Maria Barnes is involved in conceptualization, investigation, methodology, formal analysis, and writing the original draft. Haoxiang Wu is involved in methodology, validation, and writing—review and editing.

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SUPPORTING INFORMATION

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How to cite this article: Barnes NM, Wu H. Mechanisms regulating the airborne survival of *Klebsiella pneumoniae* under different relative humidity and temperature levels. *Indoor Air*. 2022;32:e12991. doi:[10.1111/ina.12991](https://doi.org/10.1111/ina.12991)